

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Uses small angle approximation for $\sin x$ or $\tan x$ Condone $y = 5 + 4x + 12x$ for this mark	AO1.1a	M1	$y = 5 + 4\sin\frac{x}{2} + 12\tan\frac{x}{3}$ $\sin x \approx x, \tan x \approx x$ $y \approx 5 + 4\left(\frac{x}{2}\right) + 12\left(\frac{x}{3}\right)$
Obtains correct equation Allow unsimplified form	AO1.1b	A1	$y \approx 6x + 5$ which is the equation of a straight line.
Concludes that the graph can be approximated by a straight line. Requires simplification of equation (condone equals) and statement.	AO2.1	R1	
			Total 3 marks

Q2.

Marking Instructions	AO	Marks	Typical Solution
Uses either $\cos x \approx 1 - \frac{1}{2}x^2$ or $\sin x \approx x$ (PI)	AO1.2	B1	$\cos 3\theta + \theta \sin 2\theta \approx 1 - \frac{(3\theta)^2}{2} + \theta(2\theta)$ $\approx 1 - \frac{5}{2}\theta^2$
Substitutes 2θ and 3θ into 'their' expression	AO1.1a	M1	
Obtains correct answer	AO1.1b	A1	
			Total 3 marks

Q3.

Marking Instructions	AO	Marks	Typical Solution
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Translates $f\left(\frac{\pi}{6}\right)$ into $\lim_{h \rightarrow 0} \frac{\sin\left(\frac{\pi}{6} + h\right) - \sin\left(\frac{\pi}{6}\right)}{h}$	AO1.1a	M1	$f\left(\frac{\pi}{6}\right) = \lim_{h \rightarrow 0} \left[\frac{\sin\left(\frac{\pi}{6} + h\right) - \sin\left(\frac{\pi}{6}\right)}{h} \right]$
Uses $\sin(A+B)$ identity to replace $\sin\left(\frac{\pi}{6} + h\right)$, to commence argument (at least two lines of argument seen)	AO2.1	M1	$= \lim_{h \rightarrow 0} \left[\frac{\sin\frac{\pi}{6} \cos h + \cos\frac{\pi}{6} \sin h - \sin\left(\frac{\pi}{6}\right)}{h} \right]$ $= \lim_{h \rightarrow 0} \left[\frac{\frac{1}{2} \cos h + \frac{\sqrt{3}}{2} \sin h - \frac{1}{2}}{h} \right]$
Obtains correct two term expression involving $\cos h$ and $\sin h$	AO2.5	M1	$= \lim_{h \rightarrow 0} \left[\frac{1}{2} \left(\frac{\cos h - 1}{h} \right) + \frac{\sqrt{3}}{2} \frac{\sin h}{h} \right]$
Deduce what happens as $h \rightarrow 0$, for one part of 'their' expression using the $\frac{\sin h}{h}$ limit of OR by using small angle approximations	AO2.2a	R1	$= \lim_{h \rightarrow 0} \left[\frac{1}{2} \left(\frac{-2\sin^2\left(\frac{h}{2}\right)}{\frac{2h}{2}} \right) + \frac{\sqrt{3}}{2} \frac{\sin h}{h} \right]$ $= \left(\lim_{h \rightarrow 0} \frac{-\sin\left(\frac{h}{2}\right)}{2} \right) \left(\lim_{h \rightarrow 0} \frac{\sin\left(\frac{h}{2}\right)}{\left(\frac{h}{2}\right)} \right) + \frac{\sqrt{3}}{2} \lim_{h \rightarrow 0} \frac{\sin h}{h}$
Deduce what happens as $h \rightarrow 0$, for the second part of 'their' expression using the limit of $(\cos h - 1)h$ OR by using small angle approximations	AO2.2a	R1	$= 0 \times 1 + \frac{\sqrt{3}}{2} \times 1$ $= \frac{\sqrt{3}}{2}$
Completes a rigorous argument leading to the correct exact value, with all the steps in the method clearly shown.	AO2.1	R1	
Total 6 marks			